

Lincoln University College

Faculty of Computer Science and Multimedia

CafeFlow - A Cafe Management System

A PROJECT REPORT

Submitted to

Department of Information Technology

Phoenix College of Management

Maitidevi, Kathmandu

In partial fulfillment of the requirements for the Bachelor in Information
Technology (BIT)

Submitted by

Amit Tharu

BIT 7th Semester

University ID: LC0003001642

2026/05/30

DECLARATION

I hereby declare that the project report entitled “**CafeFlow - A Cafe Management System**” is my original work, completed under the guidance of **Mr. Saishab Bhattarai**. This report has not been submitted elsewhere, in full or in part, for the award of any degree, diploma, or other academic recognition.

Signature:

Name: Amit Tharu

Date: 2026/05/30

LETTER OF APPROVAL



Lincoln University College

Faculty of Computer Science and Multimedia

Phoenix College of Management

Maitidevi, Kathmandu

Bachelors in Information Technology (BIT)

The Project report entitled “**CafeFlow - A Cafe Management System**” submitted by **Amit Tharu**, in partial fulfillment of the requirements for the Bachelor’s degree in Information Technology, has been accepted as a record of work independently carried out by the student.

In our opinion, it is satisfactory in the scope and quality of a project for the required degree.

Er. Santosh Dhungana

Head of Department

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ABSTRACT

CafeFlow is a web-based cafe management system designed to digitise the daily operations of small and medium cafes. Many cafes still rely on manual order taking, paper bills, and spreadsheet-based stock tracking, which leads to slow service, billing errors, stock-outs, and a lack of reliable sales insight. CafeFlow brings menu management, order taking, billing, inventory tracking, staff management, and sales analytics together in a single role-based platform. Staff can build and submit orders, advance them through a pending-preparing-completed workflow, and generate itemised bills, while administrators additionally manage the menu, monitor stock and low-stock alerts, manage the team, and view revenue and best-seller analytics derived from real order data. The system is built with Next.js, React, and TypeScript on top of a SQLite database, with secure email/password authentication and role-based access control. By replacing manual processes with a fast, consistent, and data-driven platform, CafeFlow reduces service time, improves billing accuracy, and gives cafe owners clear visibility into their business.

Keywords: Cafe Management, Order Management, Billing, Inventory, Web Application, SQLite.

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A special note of thanks goes to my friends and classmates for their constant encouragement, technical discussions, and moral support throughout this journey. Finally, I express my deepest gratitude to my family for their unwavering support, motivation, and encouragement.

CERTIFICATE

This is to certify that the project report entitled “**CafeFlow - A Cafe Management System**” submitted by **Amit Tharu, University ID: LC0003001642** to Lincoln University College in partial fulfillment of the requirements for the Bachelor in Information Technology, is a record of original work carried out under my supervision and guidance.

Supervisor Name: Mr. Saishab Bhattarai

Designation: Project Coordinator

Signature:

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BIT	Bachelor in Information Technology
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
DFD	Data Flow Diagram
ER	Entity Relationship
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
POS	Point of Sale
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language
UX	User Experience

Chapter 1:

INTRODUCTION

1.1 Introduction

A cafe handles many small but time-critical tasks every day: taking customer orders, sending them to the kitchen, tracking their progress, producing accurate bills, keeping the menu up to date, and making sure ingredients do not run out. In many small and medium cafes these tasks are still performed manually using notebooks, verbal communication, and spreadsheets. This manual approach is slow, error-prone, and offers no reliable view of sales or stock.

To address these problems, this project proposes **CafeFlow**, a web-based cafe management system that brings the core operations of a cafe into a single platform. The system allows staff to take and track orders, generate itemised bills, and view a live operational dashboard, while administrators additionally manage the menu, monitor inventory and low-stock alerts, manage staff, and analyse sales performance.

CafeFlow aims to make day-to-day cafe operations faster, more accurate, and data-driven, helping owners deliver better service and make informed business decisions.

1.2 Problem Statement

Despite the growth of the food and beverage sector, many small cafes still operate without a dedicated management system. Orders are recorded on paper and relayed verbally, which causes miscommunication, delays, and mistakes. Bills are calculated by hand, leading to errors and disputes. Stock levels are tracked informally, so ingredients run out unexpectedly. Sales data is rarely recorded in a usable form, leaving owners without insight into revenue, peak hours, or best-selling items. Existing commercial point-of-sale systems are often expensive, complex, or tied to specific hardware, making them impractical for small cafes. There is therefore a need for a simple, affordable, web-based system that unifies ordering, billing, inventory, and reporting in one place.

1.3 Objectives

1. To develop a web-based platform that digitises cafe ordering, billing, inventory, and staff management in a single application.
2. To provide secure, role-based access so that administrators and staff each see only the features relevant to their responsibilities.
3. To generate accurate, real-time sales and inventory analytics that support better business decisions.

1.4 Scope and Limitation

Scope:

1. **Menu Management:** Administrators can create, edit, delete, and toggle the availability of menu items, organised by category.
2. **Order Management:** Staff build orders from the menu and track them through a pending, preparing, and completed workflow.
3. **Billing:** The system generates itemised bills for orders, ready to be printed for the customer.
4. **Inventory Management:** Stock items are tracked with quantities, units, and low-stock thresholds, with automatic low-stock alerts.
5. **Staff Management:** Administrators manage team members and their roles.
6. **Reporting:** The system computes revenue, order counts, best-selling items, and category performance from real order data.
7. **Technology Platform:** The system is web-based, built with TypeScript and SQLite, and is accessible from browsers on desktops, tablets, and smartphones without any installation.

Limitation:

1. **Online Payment:** The current system does not integrate online or card payment gateways; payments are handled offline at the counter.
2. **Single Outlet:** The system is designed for a single cafe outlet and does not yet support multi-branch management.

3. **Internet Dependency:** As a web application, the system requires the server to be reachable over the network to operate.
4. **Kitchen Hardware:** The system does not integrate directly with kitchen display hardware or thermal printers beyond standard browser printing.

1.5 Development Methodology

The CafeFlow system follows the **Agile Methodology**, focusing on iterative development with short cycles of planning, design, implementation, and testing. The project was divided into small increments, each delivering a working feature such as authentication, menu management, order taking, or reporting. This approach allowed continuous refinement and quick adaptation to feedback. Figure 1.1 illustrates the iterative cycle followed throughout development.

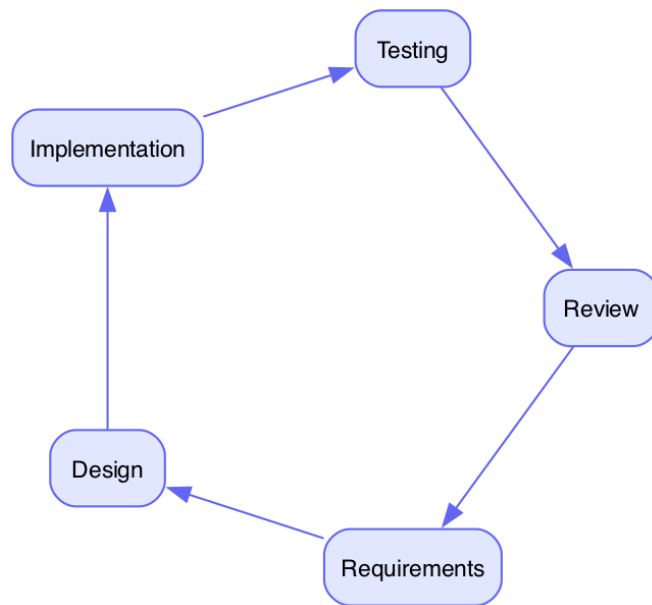


Figure 1.1: Agile / iterative development methodology

1.6 Report Organization

This report is organised into several chapters, each describing a specific area of the project:

Chapter 1: Introduction presents an overview of the project, including the problem statement, objectives, scope, limitations, and methodology.

Chapter 2: Background Study and Literature Review examines the current state of cafe operations and reviews existing systems and technologies relevant to the project.

Chapter 3: System Analysis and Design describes the requirements, feasibility study, and the models and diagrams used to design the system.

Chapter 4: Implementation and Testing explains how the design was translated into a working application and the testing carried out to verify it.

Chapter 5: Conclusion and Future Recommendations summarises the outcomes and proposes directions for future improvement.

Chapter 2:

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Cafes and coffee shops form a large and growing part of the food and beverage industry. Their operations, however, remain highly manual in many small and medium establishments. Orders are commonly taken on paper pads, prices are totalled by hand, and stock is checked by physically inspecting shelves. While these methods are simple, they do not scale well during busy periods and provide no historical record that can be analysed later.

Digital point-of-sale (POS) and restaurant management solutions exist, but they are frequently designed for larger restaurants, require dedicated hardware, or come with subscription costs that are difficult to justify for a small cafe. As a result, many cafe owners continue with manual processes despite their drawbacks.

This project studies these gaps and proposes a lightweight, browser-based alternative that runs on existing devices, requires no installation, and focuses on the specific needs of a cafe: fast ordering, accurate billing, simple inventory control, and clear sales insight.

2.2 Literature Review

Research and practice in restaurant and cafe automation consistently show that digitising order and billing processes reduces service time and human error. Point-of-sale systems are widely recognised for improving transaction accuracy and producing sales data that supports inventory and demand planning.

Modern web technologies have made such systems easier to build and deploy. Component-based front-end libraries such as React allow complex, interactive interfaces to be developed efficiently, while full-stack frameworks like Next.js unify the user interface and server-side logic in a single codebase. Lightweight embedded databases such as SQLite

provide reliable, zero-configuration data storage that is well suited to single-outlet applications.

Studies on usability in small-business software emphasise that adoption depends heavily on simplicity: systems that require minimal training and run on familiar devices are far more likely to be used consistently. CafeFlow is designed around these findings, prioritising a clean, role-based interface, fast workflows, and accessibility from any modern browser. By combining these established ideas, the project delivers a focused cafe management system that is practical for small establishments.

Chapter 3:

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

System analysis identifies what the system must do and the constraints under which it must operate. CafeFlow is analysed as a role-based web application with two types of users — **Administrator** and **Staff** — interacting with seven functional modules: authentication, dashboard, menu, orders, billing, inventory, reporting, and staff management. The analysis below captures the requirements, feasibility, and models that guided the design.

3.2 Requirement Analysis

Functional Requirements:

1. The system shall allow users to log in with an email and password and shall enforce role-based access (admin vs. staff).
2. The system shall allow administrators to create, edit, delete, and toggle availability of menu items.
3. The system shall allow staff to build an order from available menu items, set a table number and optional customer name, and submit it.
4. The system shall track each order through pending, preparing, and completed states.
5. The system shall generate an itemised bill for an order.
6. The system shall allow administrators to manage inventory items and shall raise low-stock alerts.

7. The system shall allow administrators to add and remove staff members and assign roles.
8. The system shall compute and display sales analytics (revenue, orders, top items, category split) from stored order data.

Non-Functional Requirements:

1. **Security:** Passwords shall be stored as bcrypt hashes and sessions managed with secure, HTTP-only cookies.
2. **Usability:** The interface shall be clean and responsive, usable on desktop, tablet, and mobile.
3. **Performance:** Common operations such as loading the menu or submitting an order shall complete within a fraction of a second on a local deployment.
4. **Reliability:** Data shall be persisted to a SQLite database so that it survives restarts.
5. **Maintainability:** The code shall be organised into clear layers (UI, API, data access) to ease future changes.

3.2.1 Feasibility Analysis

Technical Feasibility: The system is built with well-supported, freely available technologies (Next.js, React, TypeScript, SQLite), all of which run on standard hardware. The required skills and tools were available, making the project technically feasible.

Operational Feasibility: The system targets a real, everyday need and offers a simple, role-based interface that cafe staff can use with minimal training, so it is operationally feasible.

Economic Feasibility: The project relies entirely on open-source software and runs on commodity hardware, so development and running costs are minimal, making it economically feasible.

Schedule Feasibility: The project was divided into small Agile increments with clear milestones, allowing it to be completed within the available academic timeframe.

3.2.2 System Modelling

The system is modelled using standard UML and process diagrams. The **use case diagram** in Figure 3.1 shows the interactions between the two actors and the system,

and the **order processing flow** in Figure 3.2 shows the lifecycle of an order from creation to billing.

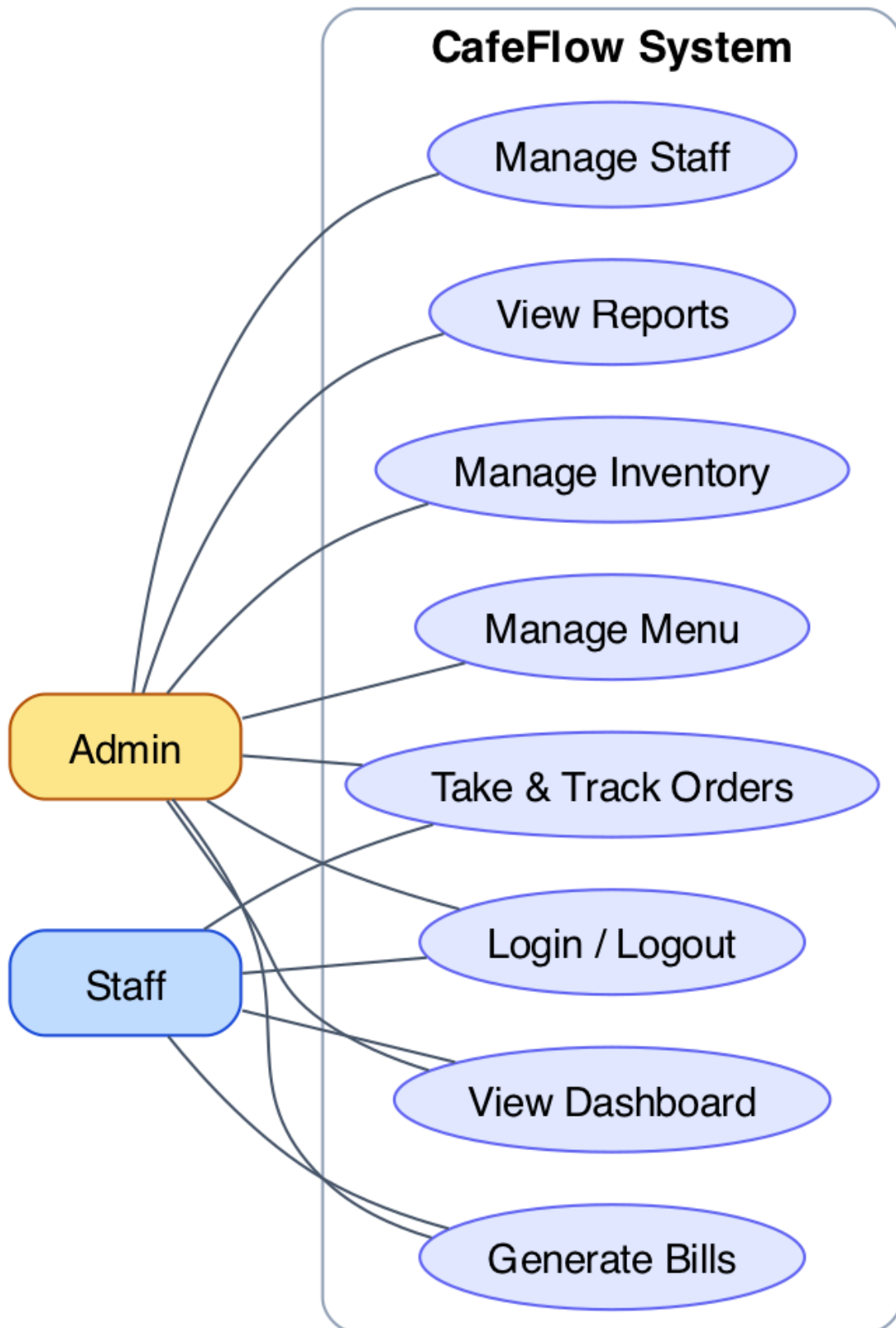


Figure 3.1: Use case diagram of CafeFlow

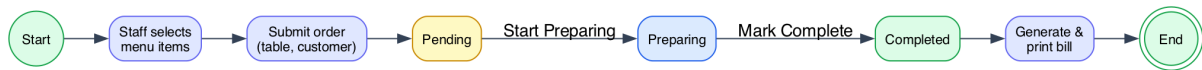


Figure 3.2: Order processing flow of CafeFlow

3.3 System Design

The system design translates the requirements and models into concrete structures: the user interface, the data model, and the runtime architecture.

3.3.1 Interface Design (UI/UX)

The interface uses a clean, role-based layout with a fixed sidebar for navigation and a content area for each module. Figure 3.3 shows the login screen and Figure 3.4 shows the administrator dashboard with live statistics and charts.

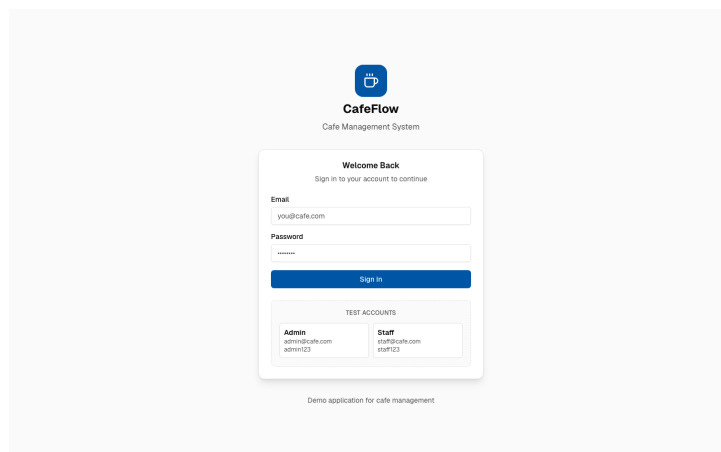


Figure 3.3: Login interface of CafeFlow

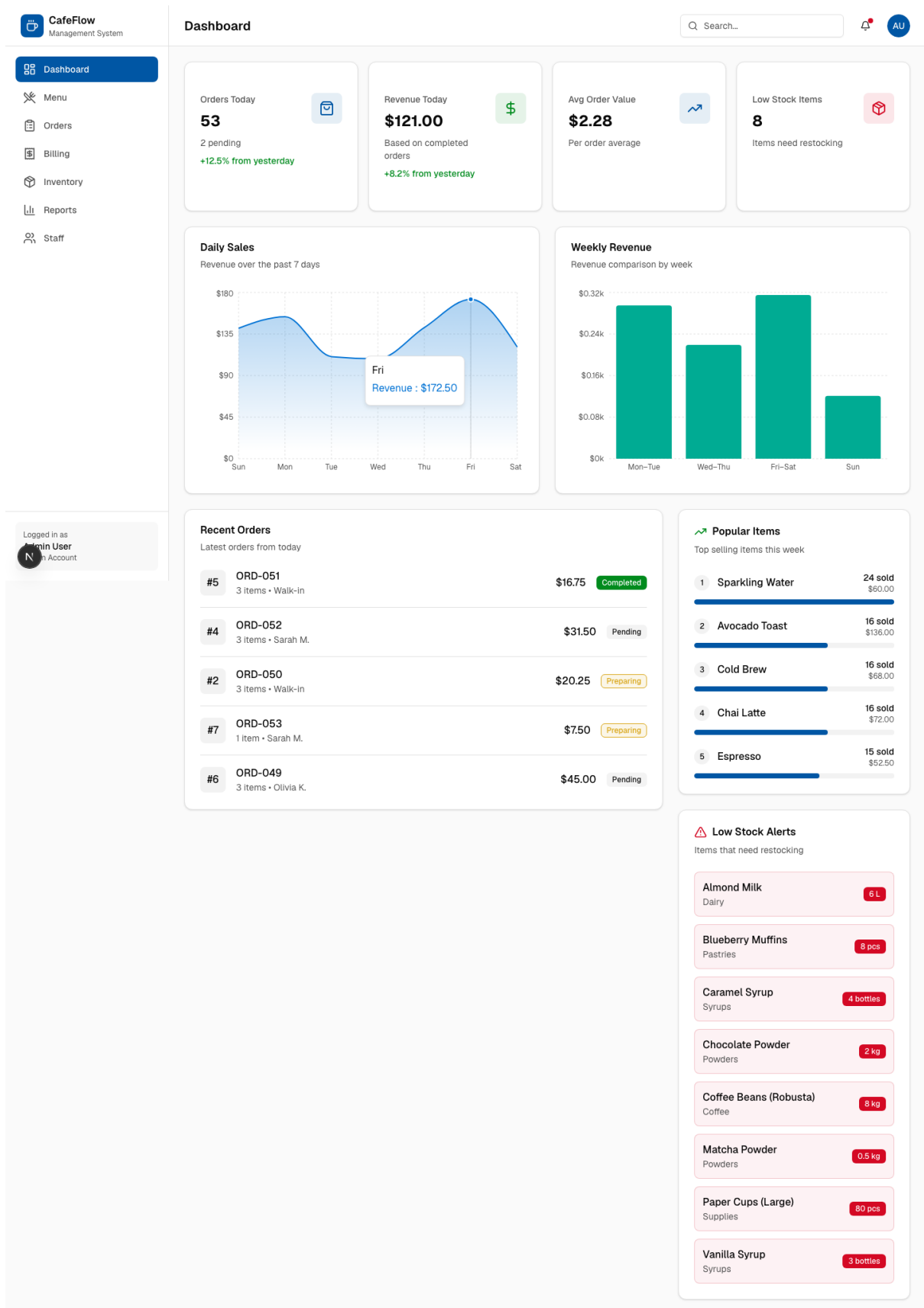


Figure 3.4: Administrator dashboard of CafeFlow

3.3.2 Refinement of Classes and Object (Data Model)

The data is stored in a relational SQLite schema. The **entity relationship diagram** in Figure 3.5 shows the main tables and their relationships: **users** and **sessions** support authentication; **menu_items** and **inventory_items** hold reference data; and **orders** and **order_items** record transactions, with each order line storing a snapshot of the item name and price so that historical bills remain correct even if the menu later changes.

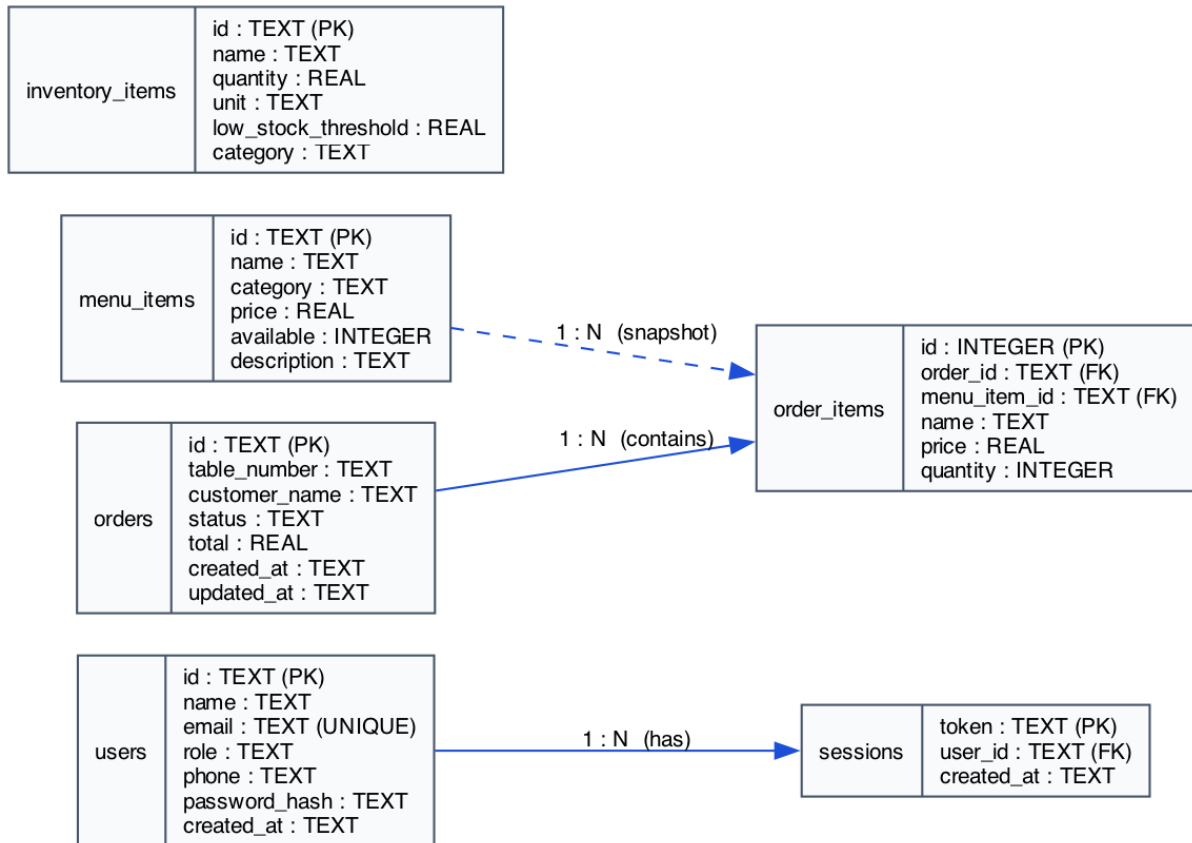


Figure 3.5: Entity relationship diagram of CafeFlow

3.3.3 Component Diagram

The system follows a layered architecture. The browser runs the React user interface and client-side state stores, which communicate over HTTP with server-side API route handlers. The handlers apply authentication and authorisation, then use a data-access layer to read and write the SQLite database. Figure 3.6 shows these components.

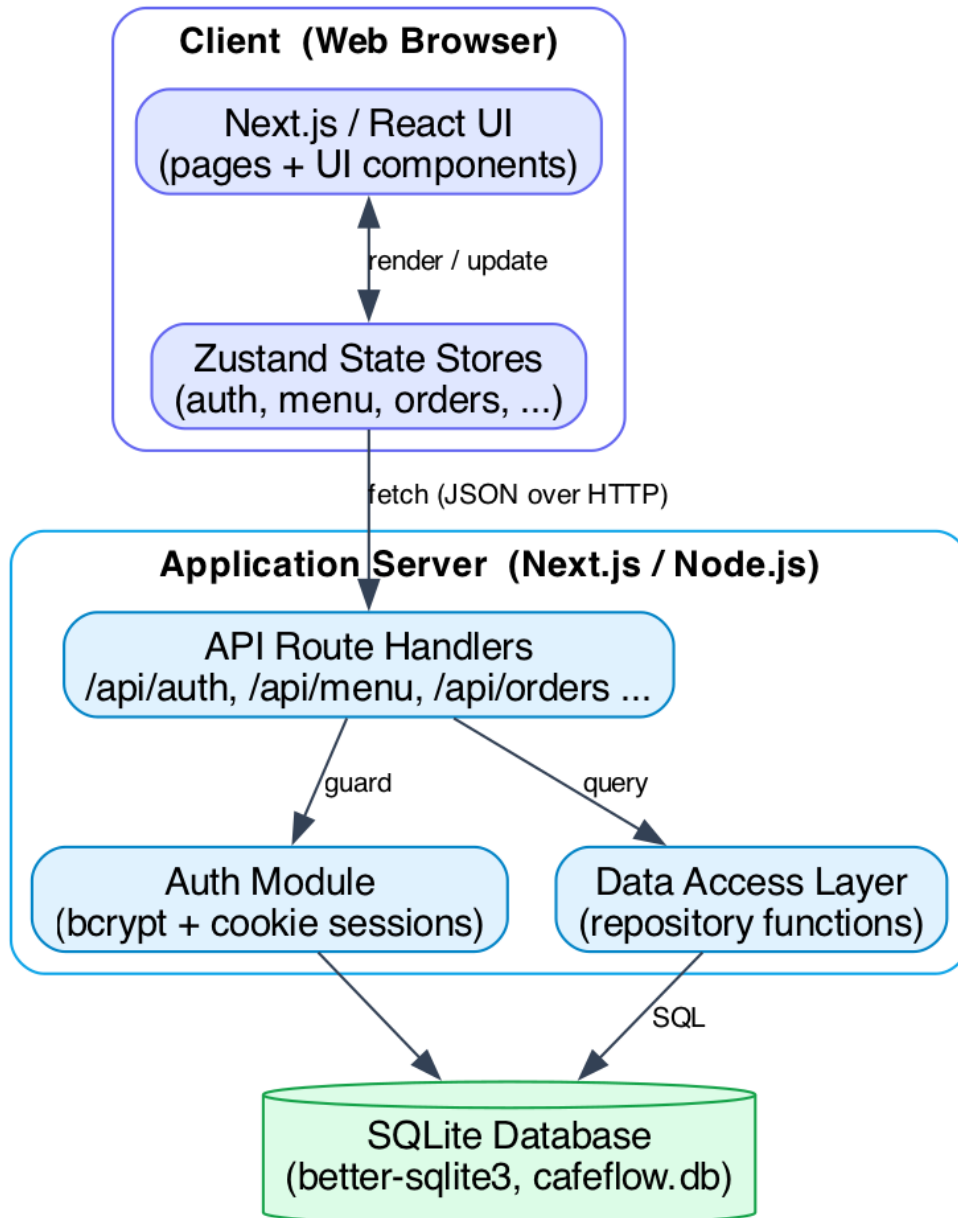


Figure 3.6: Component / architecture diagram of CafeFlow

3.3.4 Deployment Diagram

At runtime the application is deployed as a single Node.js server that serves the Next.js application and exposes the API, backed by a SQLite database file on the same host. Clients access the system through a web browser. Figure 3.7 shows the deployment.

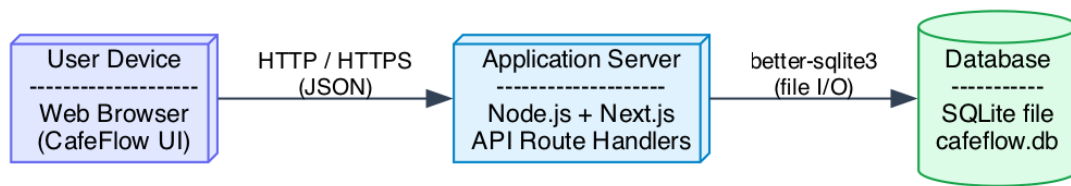


Figure 3.7: Deployment diagram of CafeFlow

Chapter 4:

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

The system was implemented using the following tools and technologies:

- **Next.js & React:** Full-stack framework and UI library for building the application and its API routes.
- **TypeScript:** Typed JavaScript for safer, more maintainable code.
- **Tailwind CSS & shadcn/ui:** Utility-first styling and accessible UI components.
- **Zustand:** Lightweight client-side state management.
- **SQLite (better-sqlite3):** Embedded relational database for persistence.
- **bcrypt:** Password hashing for secure authentication.
- **Recharts:** Charting library for the dashboard and reports.
- **Playwright:** End-to-end testing framework.
- **Visual Studio Code & Git/GitHub:** Development environment and version control.

4.1.2 Implementation Details of Modules

Authentication: Users sign in with an email and password. Passwords are verified against bcrypt hashes stored in the `users` table; on success a session token is stored in

the `sessions` table and set as a secure HTTP-only cookie. Every protected API endpoint validates the session, and administrator-only actions additionally check the user's role.

Menu: Administrators manage menu items through create, edit, delete, and availability-toggle operations, with search, category filtering, and table/card views.

Orders: Staff build an order by selecting available items, adjusting quantities, and entering a table number and optional customer name. Submitting an order stores it as `pending`; it is then advanced to `preparing` and `completed`. Each order line stores a snapshot of the item name and price.

Billing: Completed and in-progress orders can be selected to produce an itemised bill showing the cafe header, order details, line items, and total, ready for printing.

Inventory: Stock items are tracked with quantity, unit, category, and a low-stock threshold. Items at or below their threshold are flagged, and the dashboard shows low-stock alerts.

Staff: Administrators add and remove team members and assign the admin or staff role.

Reports: Revenue, order counts, average order value, best-selling items, and category performance are computed from stored order data and presented as summary cards and charts.

4.2 Testing

The system was tested using automated end-to-end tests written with Playwright, which drive a real browser through the application exactly as a user would. The tests cover every module and include both unit-level checks of individual functions and system-level checks of complete workflows. In total, 25 automated tests were executed, all of which passed.

4.2.1 Test Cases for Unit Testing

Table 4.1 lists representative unit test cases that verify individual functions and behaviours.

Table 4.1: Test Cases for Unit Testing

ID	Test Case	Expected Result	Result
U01	Login with valid admin credentials	User is authenticated and redirected to dashboard	Pass
U02	Login with an invalid password	Error message “Invalid email or password” is shown	Pass
U03	Access a protected API without a session	Request is rejected with 401 Unauthorized	Pass
U04	Add a menu item with valid data	Item is created and appears in the menu list	Pass
U05	Toggle a menu item’s availability	Availability switches and a confirmation is shown	Pass
U06	Increase an item’s quantity in the order cart	Quantity increments and the total updates	Pass
U07	Add an inventory item	Item is stored and listed in inventory	Pass
U08	Add a staff member with a duplicate email	Request is rejected with a conflict error	Pass
U09	Compute order total from line items	Total equals the sum of price $\times$ quantity	Pass
U10	Reports summary from stored orders	Revenue and order counts are calculated correctly	Pass

4.2.2 Test Cases for System Testing

Table 4.2 lists system test cases that verify complete, multi-step workflows.

Table 4.2: Test Cases for System Testing

ID	Test Case	Expected Result	Result
S01	Staff signs in and is restricted to permitted sections	Admin-only links (Menu, Reports, Staff) are hidden	Pass

ID	Test Case	Expected Result	Result
S02	Staff visits an admin-only route directly	User is redirected back to the dashboard	Pass
S03	Full menu lifecycle: add, toggle, edit, delete	Each operation succeeds and the list updates	Pass
S04	Build and submit an order	Order is created and appears under the Pending tab	Pass
S05	Advance an order pending → preparing → completed	Status updates at each step with confirmation	Pass
S06	Generate and print a bill for an order	Itemised bill is shown and print is confirmed	Pass
S07	Add, edit, and delete an inventory item	Inventory list reflects each change	Pass
S08	Add a staff member then remove with confirmation	Member is added, then removed after confirming	Pass
S09	View reports and switch analytics tabs	Summary cards and charts render for each tab	Pass
S10	Log out	Session ends and the user returns to the login page	Pass

4.2.3 Result Analysis

All 25 automated test cases passed, covering authentication and role-based access, menu, order, billing, inventory, staff, and reporting workflows. The results confirm that the system meets its functional requirements and behaves correctly across complete user journeys. The automated suite also serves as a regression guard, allowing future changes to be validated quickly. No critical defects remained at the end of testing; minor issues found during development (such as session handling and state updates) were fixed and re-verified.

Chapter 5:

CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

CafeFlow successfully delivers a complete, web-based cafe management system that unifies menu management, order taking, billing, inventory tracking, staff management, and sales reporting in a single role-based application. By replacing manual, paper-based processes with a fast and consistent digital workflow, the system reduces service time, improves billing accuracy, and provides cafe owners with clear, real-time insight into their operations. The project met all of its stated objectives and was validated by a comprehensive automated test suite.

5.2 Lesson Learnt / Outcome

The project provided valuable experience in designing and building a full-stack web application end to end. Key lessons included structuring a codebase into clear layers (interface, API, and data access), securing an application with hashed passwords and session-based authentication, modelling relational data and preserving historical accuracy through snapshots, and validating an application thoroughly with automated end-to-end tests. The work also reinforced the value of an iterative, Agile approach in delivering a working product incrementally.

5.3 Future Recommendations

While CafeFlow meets its current objectives, several enhancements are recommended for the future:

1. **Online Payments:** Integrate digital wallets and card payment gateways for cashless transactions.
2. **Multi-Outlet Support:** Extend the system to manage multiple cafe branches from a single account.
3. **Kitchen Display & Printing:** Add dedicated kitchen display screens and direct thermal-printer support.
4. **Customer-Facing Ordering:** Provide QR-code table ordering so customers can place orders directly.
5. **Advanced Analytics:** Add demand forecasting and automatic stock reorder suggestions.

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APPENDICES

The following screenshots show the main modules of the working CafeFlow application.

CafeFlow
Management System

Dashboard
Menu
Orders
Billing
Inventory
Reports
Staff

Logged in as **Admin**
2 Issues

Menu Management

Search menu items... All Categ + Add Item

Name	Category	Price	Description	Available
Americano	Coffee	\$3.75	Espresso with hot water	On
Avocado Toast	Snacks	\$8.50	Sourdough with fresh avocado	On
Bagel & Cream Cheese	Snacks	\$4.50	Toasted with cream cheese	On
Blueberry Muffin	Snacks	\$3.25	Fresh baked daily	On
Cappuccino	Coffee	\$4.50	Espresso with steamed milk foam	On
Chai Latte	Tea	\$4.50	Spiced tea with steamed milk	On
Cheesecake	Desserts	\$5.50	New York style	On
Chocolate Brownie	Desserts	\$4.00	Rich and fudgy	On
Cold Brew	Coffee	\$4.25	Slow-steeped for 12 hours	On
Croissant	Snacks	\$3.50	Buttery French pastry	On
Earl Grey	Tea	\$3.25	Black tea with bergamot	On
Espresso	Coffee	\$3.50	Rich and bold single shot	On
Fresh Orange Juice	Drinks	\$4.00	Freshly squeezed	On
Green Tea	Tea	\$3.00	Japanese sencha	On
Latte	Coffee	\$4.75	Smooth espresso with steamed milk	On
Matcha Latte	Tea	\$5.00	Premium matcha with milk	Off
Mocha	Coffee	\$5.25	Espresso with chocolate and milk	On
Smoothie Bowl	Drinks	\$7.50	Acai with granola toppings	On
Sparkling Water	Drinks	\$2.50	San Pellegrino	On
Tiramisu	Desserts	\$6.00	Classic Italian dessert	Off

Figure 5.1: Menu management

Order Management

🔔
👤 AU

Dashboard

Menu

Orders

Billing

Inventory

Reports

Staff

New Order

Pending 2

Preparing 2

Completed

All Orders

All Items

Coffee

Tea

Snacks

Drinks

Desserts

Americano \$3.75

Coffee

Avocado Toast \$8.50

Snacks

Bagel & Cream Cheese \$4.50

Snacks

Blueberry Muffin \$3.25

Snacks

Cappuccino \$4.50

Coffee

Chai Latte \$4.50

Tea

Cheesecake \$5.50

Desserts

Chocolate Brownie \$4.00

Desserts

Cold Brew \$4.25

Coffee

Croissant \$3.50

Snacks

Earl Grey \$3.25

Tea

Espresso \$3.50

Coffee

Fresh Orange Juice \$4.00

Drinks

Green Tea \$3.00

Tea

Latte \$4.75

Coffee

Mocha \$5.25

Coffee

Smoothie Bowl \$7.50

Drinks

Sparkling Water \$2.50

Drinks

🛒 Current Order

✕ Clear

Table Number *

9

Customer Name

Optional

Cappuccino \$4.50 each

- 1 + 🗑️

Croissant \$3.50 each

- 1 + 🗑️

Total: \$8.00

Submit Order

✓ Added Croissant to order

Logged in as

N

2 Issues

✕

Figure 5.2: Order builder (new order)

-  Dashboard
-  Menu
-  Orders
-  **Billing**
-  Inventory
-  Reports
-  Staff

Logged in as

N

2 Issues

X

Q Search by order ID, table, or customer...

ORD-051 completed	\$16.75
Table 5 • Walk-in	3 items
ORD-050 preparing	\$20.25
Table 2 • Walk-in	3 items
ORD-053 preparing	\$7.50
Table 7 • Sarah M.	1 items
ORD-047 completed	\$3.25
Table 9 • Olivia K.	1 items
ORD-046 completed	\$27.25
Table 14 • Sarah M.	3 items
ORD-044 completed	\$45.00
Table 1 • Walk-in	3 items
ORD-041 completed	\$7.50
Table 12 • Alex P.	1 items
ORD-048 completed	\$25.50
Table 3 • Mike R.	2 items
ORD-045 completed	\$15.50
Table 15 • Olivia K.	2 items
ORD-043 completed	\$32.00
Table 4 • Alex P.	2 items
ORD-042 completed	\$16.50
Table 7 • Liam T.	2 items
ORD-037 completed	\$19.50
Table 2 • John D.	3 items
ORD-040 completed	\$22.00
Table 15 • Sarah M.	3 items
ORD-036 completed	\$11.00
Table 6 • John D.	2 items
ORD-039 completed	\$46.50
Table 10 • Olivia K.	3 items
ORD-034 completed	\$17.75
Table 11 • Walk-in	3 items
ORD-035 completed	\$5.50
Table 6 • Liam T.	1 items



CafeFlow
123 Coffee Street, City
Tel: (555) 123-4567

Order ID:
ORD-051

Date:
May 30, 2026, 05:47 PM

Table:
5

Customer:
Walk-in

Order Items

1x Chocolate Brownie

\$4.00

1x Americano

\$3.75

2x Cappuccino

\$9.00

Subtotal

\$16.75

Tax (0%)

\$0.00

Total

\$16.75

Thank you for visiting CafeFlow!
Please come again

Added Croissant to order

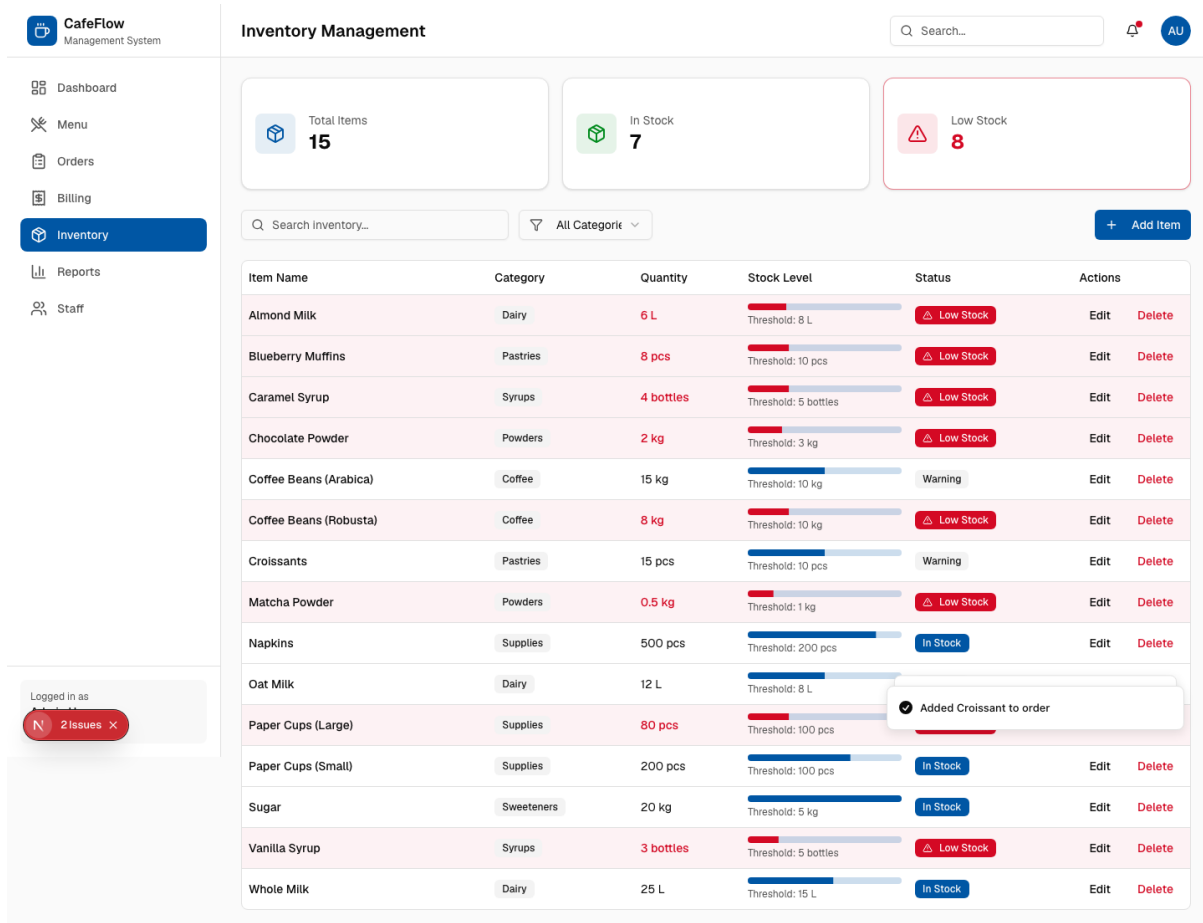


Figure 5.4: Inventory management

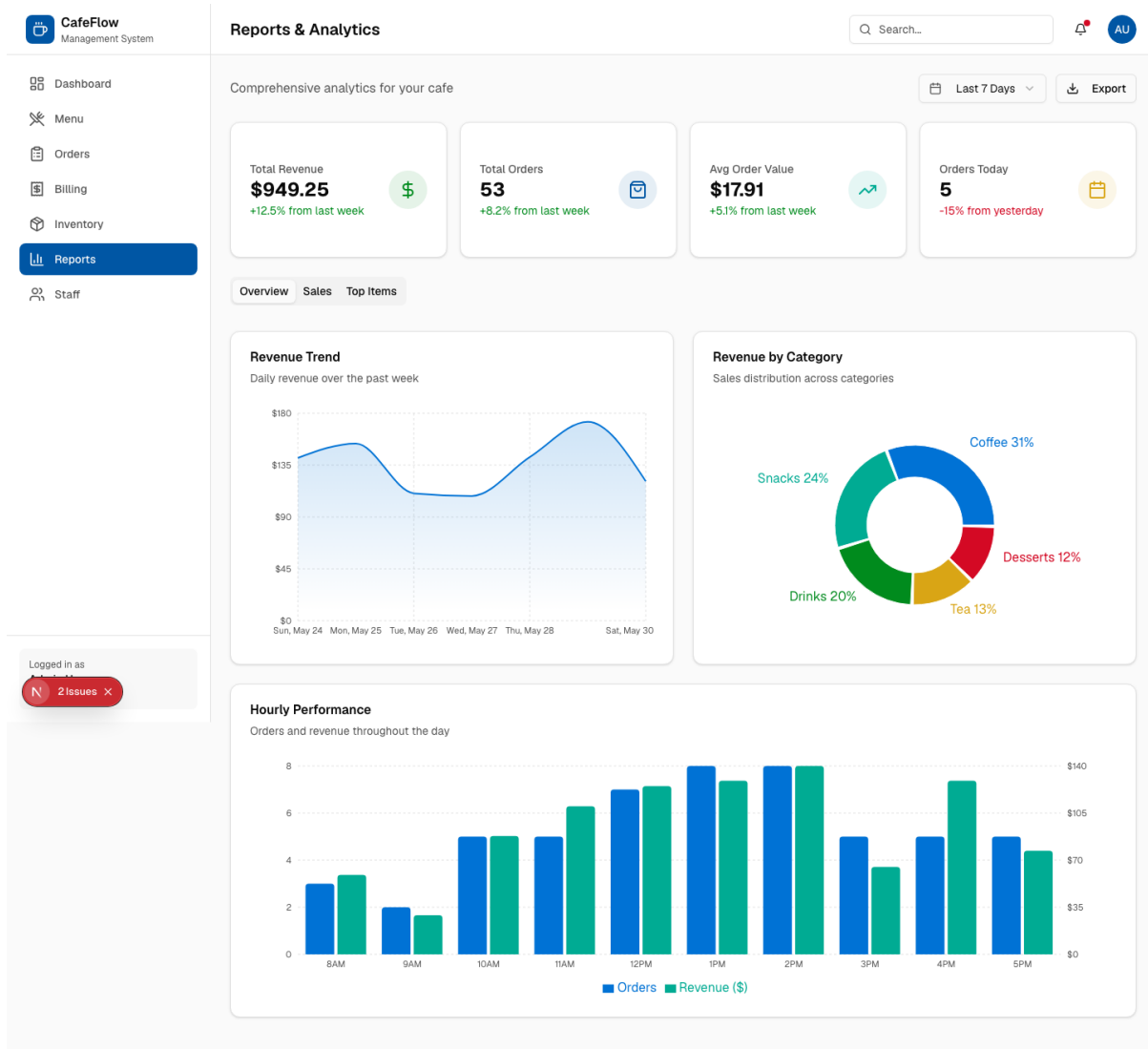


Figure 5.5: Reports and analytics

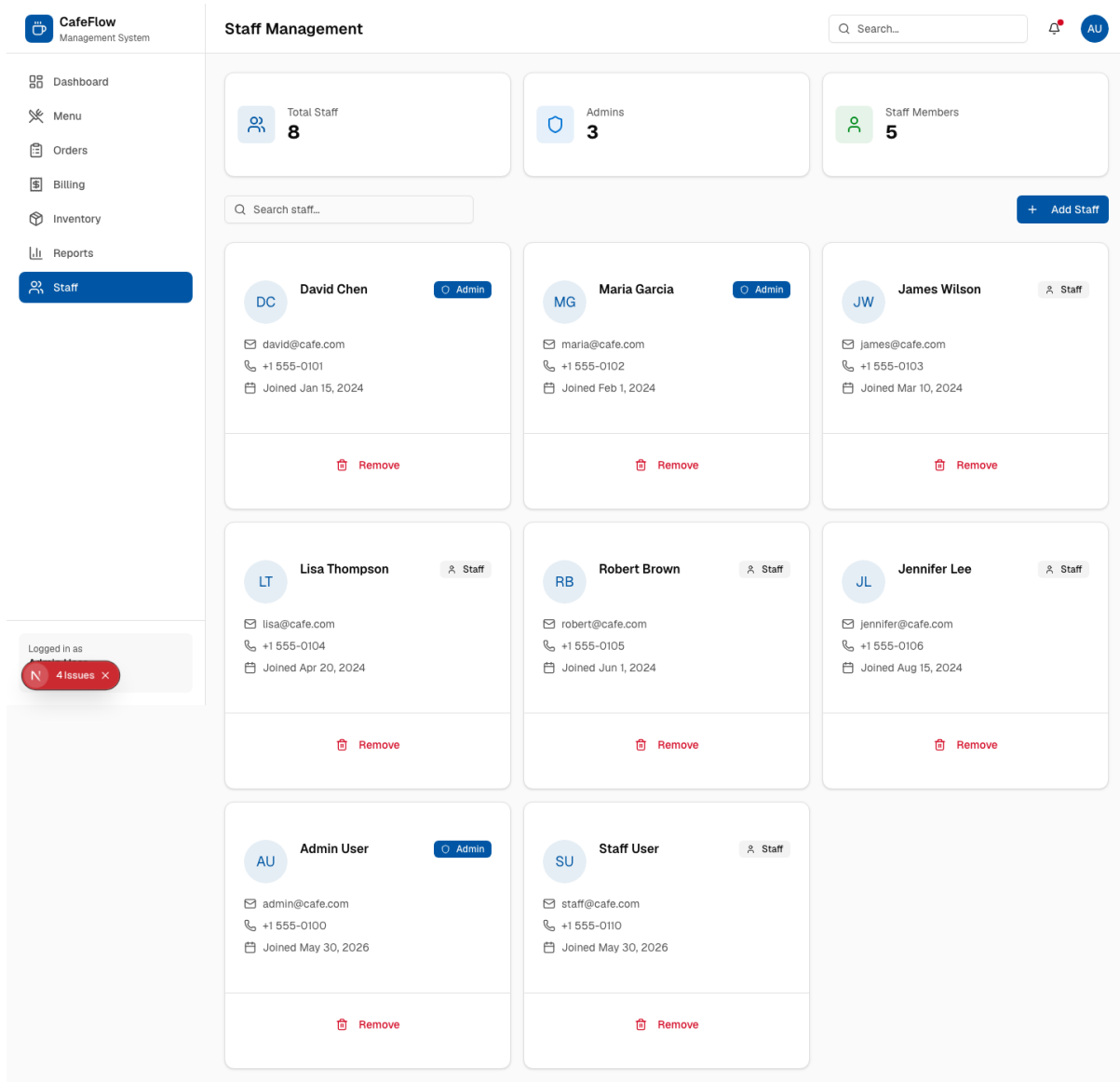


Figure 5.6: Staff management